PORTFOLIO · RESEARCH ANALYSIS

Abenezer

Mixed-Methods Research Analyst

SPSS · MAXQDA · APA 7th

Bridging rigorous methodology and clear data storytelling. Complex inferential modeling for survey datasets and systematic thematic coding for multi-case interviews, delivered as structured, defensible analysis that stands up to academic, institutional and stakeholder scrutiny.

Tools: SPSS, R, MAXQDA, ATLAS.ti, Zotero

Credentials: PhD in Educational Leadership and Policy Studies (in progress, Hawassa University); MA in Counseling Psychology

Languages: English, Amharic, Sidamu Afoo

Location: Hawassa, Ethiopia (remote collaboration)

About the samples: every table and chart in this portfolio was produced from a simulated survey dataset (N = 312) to demonstrate format, rigor and reporting standards. Real client projects remain confidential.

Services at a Glance

Three service lines, each delivered with files you keep.

1. Quantitative Analysis (SPSS)

- Data cleaning, assumption screening (normality, multicollinearity, homoscedasticity) and missing value management
- Descriptive statistics, scale reliability (Cronbach's alpha) and construct validation
- Independent and paired t-tests, ANOVA/MANOVA, chi-square, linear and logistic regression, moderation and mediation
- APA 7th summary tables, effect sizes (Cohen's d, eta squared, R^2) and high-resolution plots
- Commented, reproducible syntax files (.sps)

2. Qualitative and Thematic Synthesis (MAXQDA)

- Inductive and deductive coding of interviews, focus groups and open-ended text
- Braun & Clarke reflexive thematic analysis, grounded theory or custom frameworks
- Codebooks with operational definitions, inclusion/exclusion rules and anchor quotes
- MAXMaps and cross-case synthesis matrices; native .mx24 / .atldproj bundles

3. Academic and Institutional Documentation

- APA 7th reference auditing and structural alignment
- Methodological summaries and Results/Findings narrative drafting

Sample study used in this portfolio

Research question: which workplace factors predict teacher job satisfaction, and do they relate to retention? A simulated survey of 312 teachers measured job satisfaction (6 items), leadership support (5), workload (4) and autonomy (4) on 1–5 scales, plus department, gender, years of experience, training participation and intention to stay.

Sample	Technique	Section
1	Data cleaning and assumption screening	Quantitative
2	Reliability and construct validation	Quantitative
3	Descriptives and correlation matrix	Quantitative
4	Independent-samples t-test with Cohen's d	Quantitative
5	One-way ANOVA, Welch check, Tukey post hoc	Quantitative
6	Chi-square test of independence	Quantitative
7	Hierarchical multiple regression with diagnostics	Quantitative
8	Binary logistic regression	Quantitative
9	Moderation analysis	Quantitative
10	Mediation with bootstrapped CI	Quantitative
11–13	Reproducible syntax, thematic analysis, APA audit	Deliverables

Quantitative Work Samples (SPSS)

Each sample follows the same sequence: question, method, assumptions, results in APA 7th format, and interpretation.

SAMPLE 1 · DATA PREPARATION

Data Cleaning and Assumption Screening

Of 312 cases, 282 had complete data on the four study variables. Missingness was low (under 3% per variable) and handled by listwise deletion after a pattern check. No multivariate outliers exceeded |z| > 3.29 (0 cases flagged).

Table 1. Missing Values by Variable

Variable	Missing n	Missing %
Job satisfaction	6	1.9%
Leadership support	7	2.2%
Workload	9	2.9%
Autonomy	8	2.6%

Table 2. Distribution Screening

Variable	Skewness	Kurtosis	Shapiro–Wilk W	p
Job satisfaction	0.24	-0.45	0.981	< .001
Leadership support	-0.13	-0.34	0.990	.060
Workload	0.04	-0.14	0.994	.391
Autonomy	-0.29	-0.20	0.990	.060

Skewness and kurtosis fell within ±1, so parametric tests were retained; with n = 282 the Shapiro–Wilk test is sensitive to trivial departures.

SAMPLE 2 · MEASUREMENT

Scale Reliability and Construct Validation

Internal consistency was evaluated with Cronbach's alpha and corrected item–total correlations. All scales met the conventional .70 threshold.

Table 3. Scale Reliability

Scale	Items	Cronbach's α	Item–total r (min)	Item–total r (max)
Job satisfaction	6	0.94	0.81	0.86
Leadership support	5	0.90	0.72	0.75
Workload	4	0.79	0.56	0.63
Autonomy	4	0.84	0.67	0.69

Interpretation: alpha above .90 for satisfaction suggests possible item redundancy, which would be reviewed during scale refinement.

SAMPLE 3 · DESCRIPTIVES

Descriptive Statistics and Correlations

Table 4. Means, Standard Deviations and Range

Variable	M	SD	Min	Max
Job satisfaction	2.39	0.84	1.0	5.0
Leadership support	3.46	0.80	1.0	5.0
Workload	3.19	0.72	1.1	5.0
Autonomy	3.34	0.76	1.2	5.0

Table 5. Pearson Correlation Matrix (n = 282)

Variable	1	2	3	4
1. Job satisfaction	—			
2. Leadership support	0.52***	—		
3. Workload	-0.35***	-0.21***	—	
4. Autonomy	0.60***	0.15*	-0.09	—

** p < .05. ** p < .01. *** p < .001.*

SAMPLE 4 · GROUP COMPARISON

Independent-Samples t-Test

Question: do teachers who completed training report higher satisfaction? Levene's test supported equal variances (F = 0.13, p = .724).

Table 6. Job Satisfaction by Training Participation

Group	n	M	SD
Trained	138	2.46	0.84
Not trained	144	2.32	0.84

Result: t(280) = 1.42, p = .157, Cohen's d = 0.17, 95% CI of the difference [-0.05, 0.34]. The difference was small and not statistically significant; the interval includes zero, so the study cannot confirm a training effect.

SAMPLE 5 · MULTI-GROUP COMPARISON

One-Way ANOVA with Post Hoc Tests

Levene's test indicated unequal variances (p = .029), so Welch's F was also computed: F(2, 150.2) = 16.01, p < .001, confirming the standard result.

Table 7. Job Satisfaction by Department

Department	n	M	SD
Primary	129	2.36	0.88
Secondary	98	2.67	0.74

Department	n	M	SD
Vocational	55	1.97	0.73

Result: F(2, 279) = 13.34, p < .001, η^2 = 0.087 (medium effect).

Table 8. Tukey HSD Pairwise Comparisons

Group A	Group B	Mean difference	p (adj.)
Primary	Secondary	0.31	.013
Primary	Vocational	-0.39	.008
Secondary	Vocational	-0.70	< .001

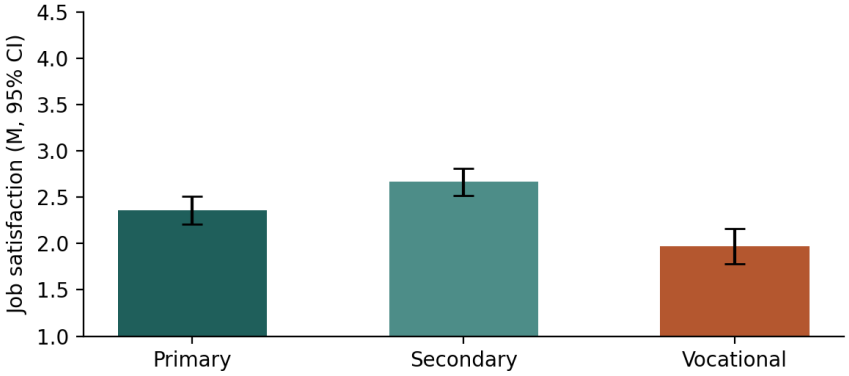


Figure 1. Mean job satisfaction by department with 95% confidence intervals.

SAMPLE 6 · CATEGORICAL ASSOCIATION

Chi-Square Test of Independence

Table 9. Intention to Stay by Gender (counts)

Gender	Not staying	Staying
Female	120	40
Male	92	30

Result: χ^2(1, N = 282) = 0.01, p = .937, Cramér's V = 0.00. Smallest expected count = 30.3, so the test assumption was met. No association was found.

SAMPLE 7 · PREDICTION

Hierarchical Multiple Regression

Block 1 entered demographic controls (gender, department, experience); Block 2 added leadership support, workload and autonomy. Block 1 explained R^2 = 0.114; adding Block 2 raised R^2 to 0.645 (ΔR^2 = 0.532, F change = 136.85, p < .001).

Table 10. Regression Predicting Job Satisfaction

Predictor	B	SE	β	t	p	VIF
Leadership support	0.42	0.04	0.40	10.04	< .001	1.06
Workload	-0.26	0.05	-0.22	-5.57	< .001	1.05
Autonomy	0.57	0.04	0.51	13.23	< .001	1.03

R^2 = 0.590, adjusted R^2 = 0.586, $F(3, 278) = 133.4$, $p < .001$; Cohen's f^2 = 1.44.

Assumption checks

Assumption	Test	Result
Multicollinearity	VIF	All VIF < 1.1 (threshold 5)
Independence of errors	Durbin–Watson	d = 1.92
Homoscedasticity	Breusch–Pagan	p = .691
Normal residuals	Shapiro–Wilk	p = .749

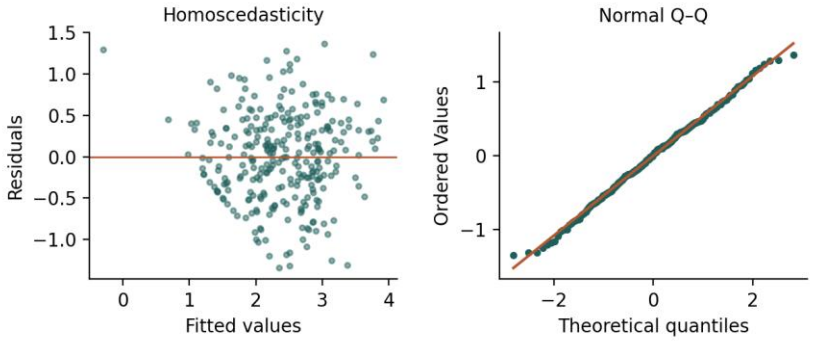


Figure 2. Residuals versus fitted values (left) and normal Q–Q plot (right).

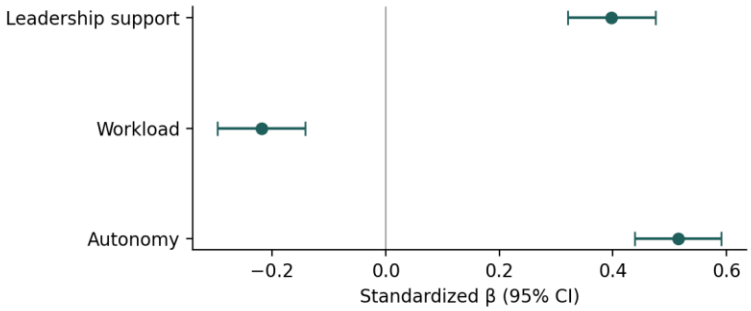


Figure 3. Standardized regression coefficients with 95% confidence intervals.

SAMPLE 8 · BINARY OUTCOME

Binary Logistic Regression

Outcome: intention to stay (1 = yes). Predictors: job satisfaction, training participation and years of experience.

Table 11. Logistic Regression Predicting Intention to Stay

Predictor	B	Odds ratio	95% CI	p
Job satisfaction	1.05	2.85	[1.95, 4.17]	< .001
Training (yes)	0.57	1.76	[0.97, 3.19]	.061
Experience (years)	-0.04	0.97	[0.92, 1.02]	.180

Model $\chi^2(3) = 40.55$, $p < .001$; Nagelkerke $R^2 = 0.199$; overall classification accuracy 77.7%.

Interpretation: each one-point increase in job satisfaction was associated with roughly 2.9 times higher odds of intending to stay, controlling for the other predictors.

SAMPLE 9 · MODERATION

Moderation Analysis

Question: does autonomy strengthen the effect of leadership support on satisfaction? Predictors were mean-centered before forming the interaction term.

Table 12. Moderated Regression

Term	B	SE	t	p
Leadership support (centered)	0.42	0.04	10.20	< .001
Autonomy (centered)	0.57	0.04	13.59	< .001
Leadership × Autonomy	0.18	0.05	3.70	< .001

R^2 = 0.609. Workload entered as a covariate.

Table 13. Simple Slopes of Leadership Support

Autonomy level	Slope
Low (−1 SD)	0.28
Mean	0.42
High (+1 SD)	0.55

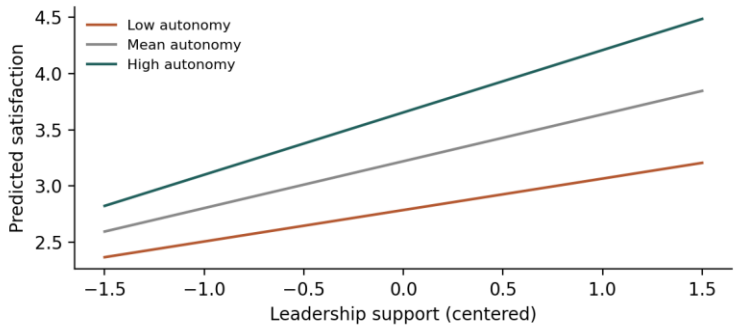


Figure 4. Interaction plot: leadership support matters more where autonomy is high.

SAMPLE 10 · MEDIATION

Mediation with Bootstrapped Confidence Intervals

Model: leadership support → autonomy → job satisfaction, with 2,000 bootstrap resamples (equivalent to PROCESS Model 4).

Table 14. Path Estimates

Path	Unstandardized B
a: Leadership → Autonomy	0.144
b: Autonomy → Satisfaction	0.585
c: Total effect	0.550
c′: Direct effect	0.465
Indirect effect (a × b)	0.084, 95% CI [0.016, 0.157]

Interpretation: the bootstrapped interval excludes zero, indicating partial mediation. Autonomy carries part of the effect of leadership support, while a direct effect remains.

Reproducible SPSS Syntax

Every quantitative delivery includes a commented .sps file so each table can be regenerated. Excerpt:

```
* ===== 1. SCREENING =====.
FREQUENCIES VARIABLES=sat lead work auto
  /STATISTICS=SKEWNESS KURTOSIS /FORMAT=NOTABLE.
EXAMINE VARIABLES=sat lead work auto /PLOT=NPPLOT.

* ===== 2. RELIABILITY =====.
RELIABILITY /VARIABLES=sat1 TO sat6
  /SCALE("Satisfaction") ALL /MODEL=ALPHA
  /SUMMARY=TOTAL.

* ===== 3. INDEPENDENT T-TEST =====.
T-TEST GROUPS=train(0 1) /VARIABLES=sat
  /CRITERIA=CI(.95).

* ===== 4. ONE-WAY ANOVA =====.
ONEWAY sat BY dept /STATISTICS=DESCRIPTIVES HOMOGENEITY WELCH
  /POSTHOC=TUKEY ALPHA(0.05).

* ===== 5. HIERARCHICAL REGRESSION =====.
REGRESSION /STATISTICS COEFF R ANOVA CHANGE COLLIN TOL
  /DEPENDENT sat
  /METHOD=ENTER gender dept exp
  /METHOD=ENTER lead work auto
  /RESIDUALS DURBIN /SCATTERPLOT=(*ZRESID,*ZPRED).

* ===== 6. LOGISTIC REGRESSION =====.
LOGISTIC REGRESSION VARIABLES stay
  /METHOD=ENTER sat train exp /PRINT=CI(95).
```

Delivery checklist (quantitative projects)

- Cleaned dataset (.sav) with variable and value labels
- Commented syntax (.sps) and output file (.spv)
- APA 7th tables and figures at publication resolution
- Assumption-check log with decisions explained
- Results narrative ready to paste into your manuscript or report

Qualitative Sample (MAXQDA)

SAMPLE 12 · THEMATIC ANALYSIS

Interview Study: Leaders' Experience of Institutional Change

Reflexive thematic analysis (Braun & Clarke) of 14 semi-structured interviews across three cases. Coding moved from an inductive first pass to deductive refinement against a theoretical framework, with a memo trail for each decision.

Codebook excerpt

Theme	Definition	Include when
Trust as currency	Change succeeds when relational trust exists	Speaker names trust, credibility or betrayal
Resource tension	Ambition constrained by limited capacity	Speaker cites time, funds or staffing limits

> *"We could change the policy overnight, but not the people's confidence."*

Illustrative anchor quote. Deliverables: codebook, MAXMaps, cross-case matrix and native .mx24 project.

Mixed-methods joint display

Quantitative result	Qualitative theme	Integration
Leadership support strongest predictor	Trust as currency	Confirms
Autonomy mediates leadership effect	Resource tension	Explains

Documentation and Process

APA 7th Reference Audit and Results Drafting

- Citation-to-reference cross-check, DOIs, hanging indents and capitalization
- Statistical reporting aligned with APA 7th (symbols, italics, decimal places)
- Methodological summary and Results section drafted from the final output tables

How we work

Step	What happens
1. Scope	Review your dataset, research questions and deadline
2. Plan	Agree on the analysis plan and milestones
3. Analyze	Full assumption checks, or systematic coding with an audit trail
4. Deliver	Tables, narrative, syntax and native project files, with regular progress updates

Let's discuss your dataset, research questions and upcoming deadlines.

Send an invitation or message on Upwork to get started.